

Time series forecasting

Generic supervised learning approach:

samples are i.i.d. $\rightarrow$ not connected to each other.

{ samples before time t .

$\rightarrow$ sample at time t can use previous samples to understand the history and come up better prediction.

Bike rental dataset on kaggle.

Forecasting problem:

$$\hat{y}_{t+h|t} = f(y_t, y_{t-1}, \dots, x_t, x_{t-1}, \dots)$$

h = horizon (how far ahead we predict)

t = forecast origin



frequency! spacing b/w the samples.

Information set! Dataset (target and features)

forecasting contract:

$$\hat{y}_{t+h|t} = f(I_t)$$

① Trend: ^{$\rightarrow T_t$} sustained long-term inc/rease/decrease.

② seasonality: S_t A pattern tied to known and fixed period m. can be weekly, monthly, yearly.

③ cycle: A rise and fall without a fixed calendar period.

④ Remainder: R_t what remains after trend and seasonal pattern is removed.
residual

$$y_t = T_t + S_t + R_t$$

Additive model.

$$y_t = T_t \times S_t \times R_t$$

multiplicative models

* Baseline:

① mean:

$$\hat{y}_{t+h|t} = \bar{y}_t$$

② Naive:

$$\hat{y}_{t+h|t} = y_t$$

③ moving average:

$$\hat{y}_{t+1|t} = \frac{y_{t-1} + y_t}{2}$$

$$= \frac{1}{w} \sum_{j=0}^J y_{t-j}$$

④ Drift:

$$\hat{y}_{t+h|t} = y_t + h \cdot \frac{y_t - y_1}{t-1}$$

Extends the avg. change from first to latest observation.

⑤ Seasoning:

$$\hat{y}_{t+h|t} = y_{t+h} - m(k+1)$$

Repeats the matching position from the latest observation with season, $m(k+1)$

* Evaluate:

$$\underline{RMSE:} \quad \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

$$\underline{MAPE:} \quad \frac{\sum_{i=1}^n |y_i - \hat{y}_i|}{|y|} \times 100$$

$$MAE: \sum_{i=1}^n |y_i - \hat{y}_i|$$

$$RMSLE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\log(y_i + 1) - \log(\hat{y}_i + 1))^2}$$

* Lag: Lag aligns today's target values that were already known.

Supervised framing.

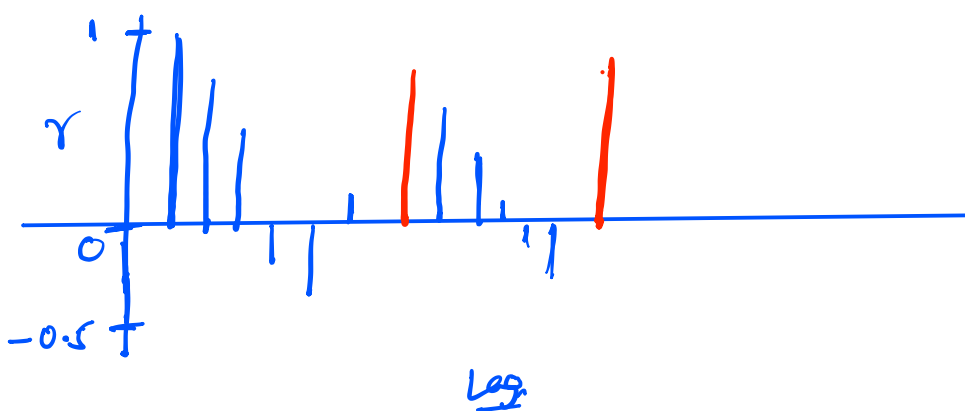
$$x_t = [y_{t-1}, y_{t-2}, \dots, y_{t-p}]. \text{ target} = y_t$$

Time	Lag 1	Lag 7	Last 7-day mean	target
Day t	y_{t-1}	y_{t-7}	$\frac{1}{7} \sum_{j=1}^7 y_{t-j}$	y_t

ACF $\Rightarrow$ Auto correlation function:

How strongly this series resembles delayed version of itself.

$$\gamma_p = \frac{\sum_{t=p+1}^T (y_t - \bar{y})(y_{t-p} - \bar{y})}{\sum_{t=1}^T (y_t - \bar{y})^2}$$



Input $\rightarrow$ Hi, how are you? ?

lapt $\rightarrow$ most probable next word. $\rightarrow$ Hi

Input $\rightarrow$ Hi, how are you? Hi

* stationarity:

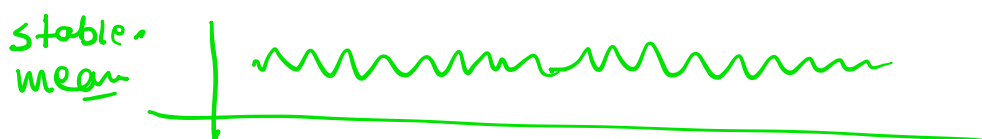
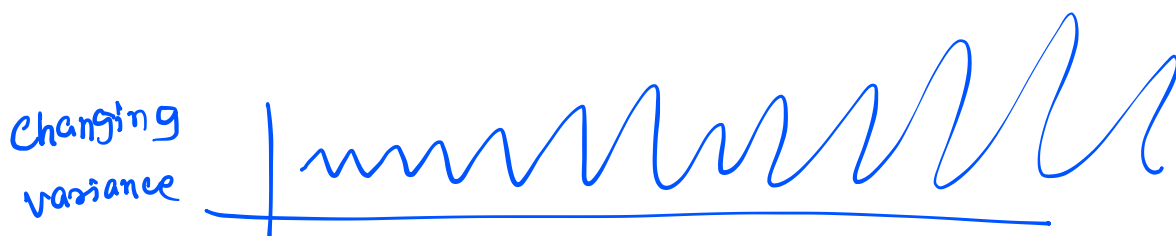
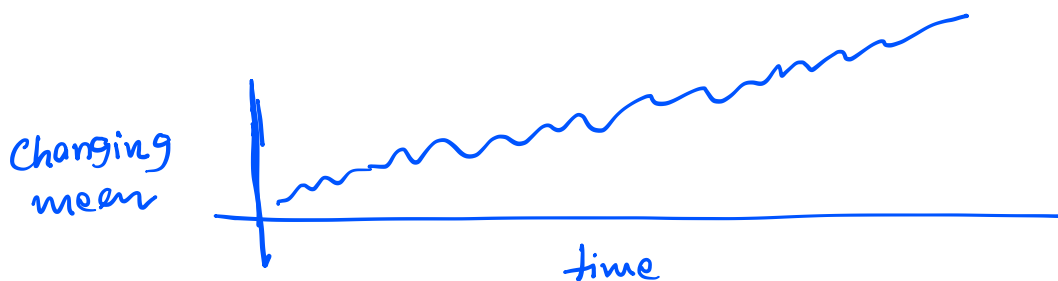
whether the statistical behaviour in one period remains a useful descriptor for another period

Stationary looking data:

98, 103, 100, 96, 102, 101, 99, ...

non-stationary looking data:

100, 108, 115, 130, 134, ...



① stable mean: $E(Y_t) = \mu$

② stable variance: $\text{var}(Y_t) = \sigma^2$

Stationarity says its statistical behaviour is stable,
not that its future values are known.

gt helps: ① model generalization

② ACF are interpretable.

How to find stationarity:

① plot the data over time.

② compare time windows.

③ Augmented - Dickey - Fuller test
→ P value.

* Differencing:

$$\Delta Y_t = Y_t - Y_{t-1}$$

↓
↓
 today yesterday

100, 108, 116, 124, 132, ...

First difference:

$$\Delta Y_t = Y_t - Y_{t-1} = (1 - B)Y_t$$

$B \rightarrow$ backshift operator

$$\boxed{BY_t = Y_{t-1}}$$

Seasonal difference:

$$\Delta_m Y_t = Y_t - Y_{t-m} = (1 - B^m)Y_t$$

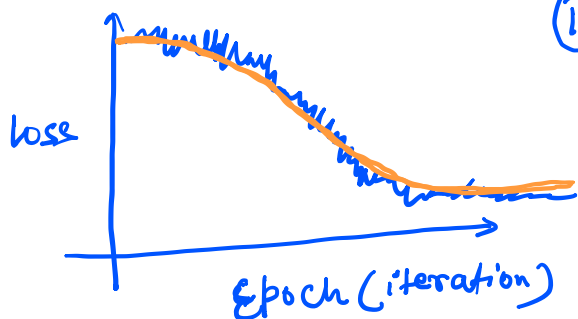
$m =$ period

Try to forecast changes, not the actual values.
 $\downarrow$
differences

Smoothing:

Tricks to reduce the effect of short-term noise.

100, 104, 98, 103, 97, 105, 101



① moving average:

3-day MA:

$$M_t = \frac{Y_t + Y_{t-1} + Y_{t-2}}{3}$$

Reduces noise.

Limitation:

- ① Fluctuations are hard to capture.
- ② All points are equally weighted.

* Exponential Smoothing:

↗ level

$$l_t = \alpha Y_t + (1-\alpha) l_{t-1}$$

Y_t = new observation

l_{t-1} = previous level estimate

α = smoothing parameter, $0 \leq \alpha \leq 1$

$$l_0 = Y_0$$

$$l_1 = \alpha Y_1 + (1-\alpha) l_0$$

$$l_2 = \alpha Y_2 + (1-\alpha) l_1$$

new portion of portion of
belief new evidence old belief.

$$l_{t-1} = \underline{100}, Y_t = \underline{120}, \alpha = \underline{0.3}$$

$$\begin{aligned} l_t &= 0.3 \times 120 + 0.7 \times 100 \\ &= 36 + 70 \\ &= \underline{\underline{106}} \end{aligned}$$

if $\alpha = 0.1 \rightarrow$ Trusts history more, reacts slowly,
less sensitive to noise.

if $\alpha = 0.9 \rightarrow$ opposite of above

Why is it called Exponential?

$$L_t = \alpha Y_t + (1-\alpha) L_{t-1}$$

$$L_{t-1} = \alpha Y_{t-1} + (1-\alpha) L_{t-2}$$

$$L_t = \alpha Y_t + \alpha(1-\alpha) Y_{t-1} + \alpha(1-\alpha)^2 Y_{t-2}$$

$$\alpha, \alpha(1-\alpha), \alpha(1-\alpha)^2, \dots$$

Decline is exponential,

* forecasting using SES!

trend and

Yes we can, but a seasonality is missing,

Let's add trend,

L_t = current level

b_t = current trend

$$L_t = \alpha Y_t + (1-\alpha)(L_{t-1} + b_{t-1})$$

$$b_t = \beta (L_t - L_{t-1}) + (1-\beta) b_{t-1}$$

$$b_0 = (Y_1 - Y_0)$$

Holt's method:

$$\hat{Y}_{t+h|t} = L_t + h b_t$$

$$0 \leq \beta \leq 1$$

Holt's - winter's method!

Add Seasonality!

l_t = level

b_t = trend

S_t = seasonal effect

Remove the known seasonal effect

$$l_t = \alpha (Y_t - S_{t-m}) + (1-\alpha) (l_{t-1} + b_{t-1})$$

$$S_t = \gamma (Y_t - l_{t-1} - b_{t-1}) + (1-\gamma) S_{t-m}$$

m = period

$$\hat{Y}_{t+h|t} = l_t + h \cdot b_t + S_t$$

ARIMA!

models a stationary version of the series using its past values and past forecast errors.

$$ARIMA(p, d, q)$$

AR $\rightarrow$ Auto Regressive $\rightarrow$ using past to predict the current

I $\rightarrow$ Integrated $\rightarrow$ difference of series.

MA $\rightarrow$ moving average $\rightarrow$ using prev. forecast error to predict the current

$\left\{ \begin{array}{l} p \rightarrow \text{no. of Autoregressive lags} \\ d \rightarrow \text{no. of differences} \\ q \rightarrow \text{no. of lagged errors,} \end{array} \right.$

① AR part:

Auto $\rightarrow$ self

Regression $\rightarrow$ predicting values using past values

$$Y_t = C + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + \epsilon_t$$

$p=1$ (one lag)

$$Y_t = C + \phi_1 Y_{t-1} + \epsilon_t \rightarrow \text{AR}(1)$$

if ϕ is close to 1 $\rightarrow$ fluctuation persist for a long time

if $\phi \approx 0 \rightarrow$ past has little influence

if $\phi < 0 \rightarrow$ Alternate around mean.

Desire $\rightarrow \phi < 1$

② I part:

Random walk:

$$Y_t = Y_{t-1} + \epsilon_t$$

$$\Delta Y_t = Y_t - Y_{t-1} = \epsilon_t$$

① Diff. the series before modeling

② Forecast the differences.

$d=0 \rightarrow$ no difference,

$d=1 \rightarrow$ first difference

$d=2 \rightarrow$ second difference

$AR(1,1) \rightarrow$ 1 lag and 1 difference,

$$\Delta Y_t = Y_t - Y_{t-1}$$

$$\Delta^2 Y_t = \Delta Y_t - \Delta Y_{t-1}$$

ARIMA model predicts:- $\hat{\Delta Y}_{t+1} = 8$

$$Y_t = 130$$

$$\hat{Y}_{t+1|t} = 130 + 8 = \underline{138}.$$

Part 3: MA :

MA model, uses prev. forecast errors.

$$Y_t = C + E_t + \theta_1 E_{t-1} + \dots + \theta_q E_{t-q}$$

$MA(1) \Rightarrow$

$$Y_t = C + E_t + \theta E_{t-1}$$

ϵ_t is unknown but previous errors are known.

$$Y_t = 100 + \overset{\rightarrow c}{0.6} \overset{\rightarrow \theta}{\epsilon_{t-1}} + \epsilon_t$$

let's assume $\epsilon_{t-1} = 10$

$$\hat{Y}_t = 100 + 0.6(10)$$
$$= 100 + 6 = 106$$

Complete ARIMA model:

Differenced series:

$$Z_t = (1-B)^d Y_t$$

B is the backshift operator:

$$B Y_t = Y_{t-1}$$

$$\underbrace{Z_t}_{\text{Differenced}} = c + \underbrace{\sum_{i=1}^P \phi_i Z_{t-i}}_{\text{AR}} + \epsilon_t + \underbrace{\sum_{j=1}^Q \theta_j \epsilon_{t-j}}_{\text{MA}}$$

$\rightarrow$ Differenced

ARIMA(1, 1, 1)

$$\Delta Y_t = c + \phi \Delta Y_{t-1} + \theta \cdot \epsilon_{t-1} + \epsilon_t$$

$$c = 2, \phi = 0.6, \theta = 0.4$$

$$\Delta Y_t = 10$$

$$\epsilon_t = -3$$

$$\begin{aligned}\hat{\Delta Y_{t+1}} &= 2 + 0.6 \times 10 + 0.4 \times (-3) \\ &= 6.8\end{aligned}$$

if the current level was 150

$$\hat{Y_{t+1}|t} = 150 + 6.8 = 156.8$$